EOCIS_LST_SLSTR: Quality Processes Review

Author(s): ***Darren Ghent***
University of Leicester

Reviewer(s): ***Samantha Malone, Sam Hunt***
NPL

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1 Introduction

1.1 Scope

This document summarises a Data Product Quality Processes Review for *the EOCIS: Daily land Surface Temperature from SLSTR (Sea and Land Surface Temperature Radiometer) on Sentinel 3x, level 3 collated (L3C) global product, version 4.00* as part of the EODH Data Product Quality Process [AD-1]. The purpose is to demonstrate that the product has been developed and generated following processes that adhere to best practice to an extent that is adequate for its intended use.

To achieve this, the EODH Data Product QA Process adheres to the ESA/NASA EO Product QA Framework [RD-1]. This framework defines requirements for a variety of aspects of Data Product quality – awarding grades of *Basic*, *Good*, *Excellent*, or *Ideal*. Such assessments are reported with a maturity matrix – this is presented in Section 2. The rest of the document provides the underpinning evidence justifying the awarded grades. This assessment has been completed by University of Leicester as a self-assessment and reviewed by the EODH QA Service.

The result of this evaluation, referred to as a *QA Check Result*, will be added as an annotation to the EODH Catalogue for user interaction.

1.2 Changes Log

Issue	Date	Description
1.0	18/05/2026	Draft for pre-publication review.

1.3 Applicable Documents

[AD-1] EODH QA Service, 'Data Product Quality Assurance Process', 2025.

1.4 Reference Documents

[RD-1] Hunt et al., 'Earth Observation Mission Quality Assessment Framework' v2.2, EDAP Project Document, 2021. Available Online.

1.5 Acronyms & Abbreviations

EDAP	Earthnet Data Assessment Project
EO	Earth Observation
EODH	Earth Observation Data Hub
ESA	European Space Agency
NASA	National Aeronautics and Space Administration
NCEO	National Centre for Earth Observation
NPL	National Physical Laboratory
QA	Quality Assurance

2 Review Summary – Product Maturity Matrix

The *Data Product Quality Processes Review* is aimed at demonstrating that a given Data Product is developed and generated following processes that adhere to best practice to an extent that is adequate for its intended use.

To achieve this, the EODH *Data Product QA Process* adheres to the ESA/NASA EO Product QA Framework [RD-1]. Such assessments are reported in a maturity matrix style, where each box represents an area of assessment for which underpinning requirements are defined – the box colour represents the achieved grade, *Basic*, *Good*, *Excellent*, or *Ideal*. Additionally:

- *Not Assessed* means this aspect of data quality was not assessed at this time.
- Not Assessable means the assessment of this aspect of data quality was not possible due to a lack of available evidence.
- The padlock symbol means that the supporting evidence reviewed is not publicly available.

The Data Product Quality Processes Review for the EOCIS: Daily land Surface Temperature from SLSTR (Sea and Land Surface Temperature Radiometer) on Sentinel 3x, level 3 collated (L3C) global product, version 4.00 product is presented in

Figure 1.

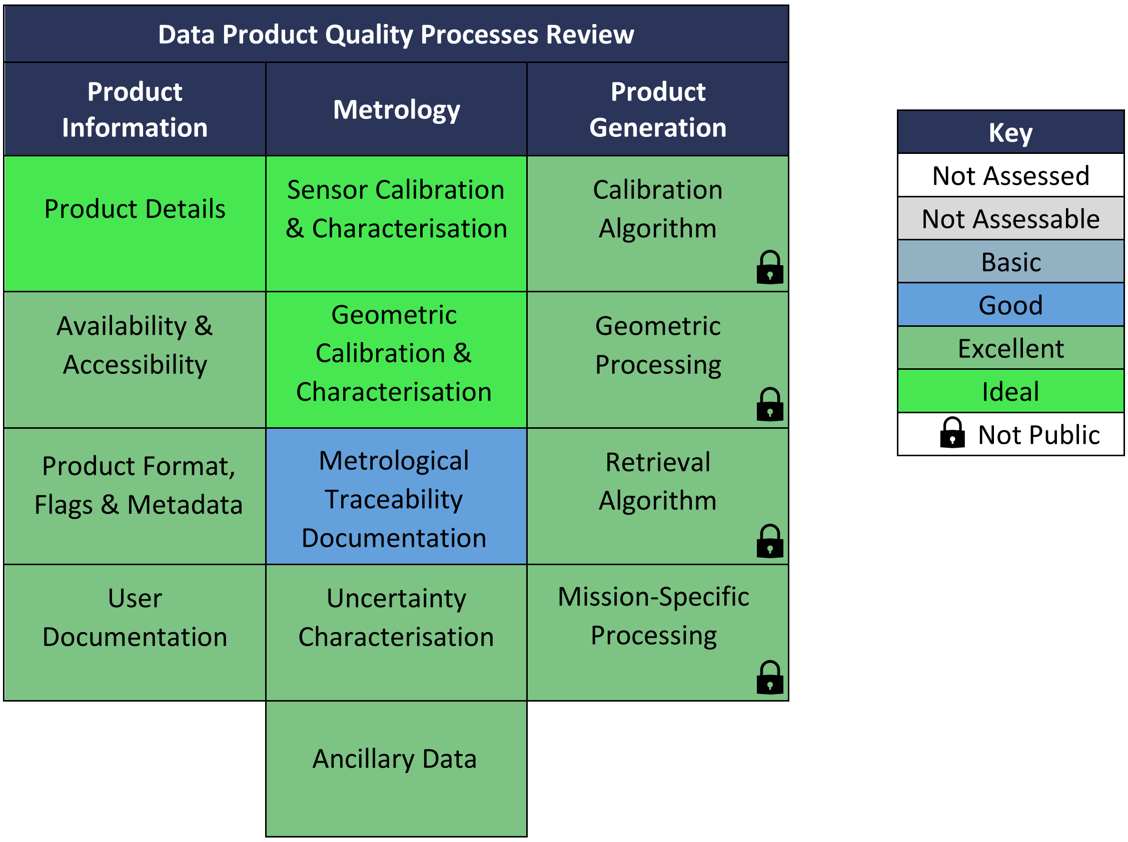

Figure 1 - Data Product Quality Processes Review Maturity Matrix for EOCIS: Daily land Surface Temperature from SLSTR (Sea and Land Surface Temperature Radiometer) on Sentinel 3x, level 3 collated (L3C) global product, version 4.00

The rest of the document provides the underpinning evidence justifying the awarded grades, as follows:

- Section 3 provides the evidence for the Product Information column.
- Section 4 provides the evidence for the Metrology column.
- Section 5 provides the evidence for the Product Generation column.

3 Product Information

Product Details	
Grade: Ideal	
Justification	*Open access Global Climate Data Record at high spatial resolution which meets GCOS Requirements on measurement uncertainty and stability*
Product Name	*EOCIS: Daily land surface temperature from SLSTR (Sea and Land Surface Temperature Radiometer) on Sentinel 3x, level 3 collated (L3C) global product, version 4.00*
Sensor Name	*SLSTR*
Sensor Type	*Thermal – radiometer*
Mission Type	*Constellation – 2 satellites*
Mission Orbit	*Sun Synchronous*
Product Version Number	*V4.00*
Product ID	*EOCIS_LST_SLSTR*
Processing level of product	*Level 3C*
Measured Quantity Name	*Land surface temperature*
Measured Quantity Units	*K*
Stated Measurement Quality	*1 K accuracy*
Spatial Resolution	*0.01°*
Spatial Coverage	*Global*
Temporal Resolution	Daily (Day / Night)
Temporal Coverage	2016 – 2024
Point of Contact	Darren Ghent, djg20@leicester.ac.uk
Product locator (DOI/URL)	SLSTR-A: https://dx.doi.org/10.5285/a784eeb9287b43bcb63ccae59e6af82e SLSTR-B: https://dx.doi.org/10.5285/fc0bc3d5887d441296091a8025f8f45d
Conditions for access and use	Open Access
Limitations on public access	cc-by-4.0
Product Abstract	This dataset contains land surface temperatures (LSTs) and their uncertainty estimates from the Sea and Land Surface Temperature Radiometer (SLSTR) on Sentinel 3A and Sentinel 3B. Satellite land surface temperatures are skin temperatures, which means, for example, the temperature of the ground surface in bare soil areas, the temperature of the canopy over forests, and a mix of the soil and leaf temperature over sparse vegetation. The skin temperature is an important variable when considering surface fluxes of, for instance, heat and water.

	Daytime and night-time temperatures are provided in separate files corresponding to the morning and evening Sentinel 3B equator crossing times which are 10:00 and 22:00 local solar time. Per pixel uncertainty estimates are given in two forms, first, an estimate of the total uncertainty for the pixel and second, a breakdown of the uncertainty into components by correlation length. Also provided in the files, on a per pixel basis, are the observation time, the satellite viewing and solar geometry angles, a quality flag, and land cover class. The dataset coverage is global over the land surface. LSTs are provided on a global equal angle grid at a resolution of 0.01° longitude and 0.01° latitude. SLSTRx achieves full Earth coverage in 1 day so the daily files have gaps where the surface is not covered by the satellite swath during day or night on that day. Furthermore, LSTs are not produced where clouds are present since under these circumstances the IR radiometer observes the cloud top which is usually much colder than the surface. Dataset coverage starts on 1st May 2026 for Sentinel 3A and 17th November 2018 for Sentinel 3B. Both continue until 31st December 2024. There are minor interruptions (1-10 days) during satellite/instrument maintenance periods or instrument anomalies. The dataset was produced by the University of Leicester (UoL) and LSTs were retrieved using the (UoL) LST retrieval algorithm and data were processed in the UoL processing chain. The dataset was produced as part of the UK Earth Observation Climate Information Service (EOCIS) and is based on development funded under ESA CCI with additional funding from NCEO.

Availability & Accessibility	
Grade: Excellent	
Justification	*Open Access compliant with FAIR principles*
Compliant with FAIR principles	*Yes*
Data Management Plan	*No*
Availability Status	*Open Access*

Product Format, Flags & Metadata	
Grade: Excellent	
Justification	*NetCDF CF-compliant, but lacks CEOS ARD compliance.* Comprehensive set of metadata and data flags.
Product File Format	*netCDF4*
Metadata Conventions	*CF-1.8*
Analysis Ready Data?	*No*

User Documentation		
Grade: Excellent		
Justification	*Detailed documentation for both use and technical details*	
Document	*Reference*	*QA4ECV Compliant*
Product User Guide	*Ghent, 2024, EOCIS Quick Start Guide: Land Surface Temperature Interim Climate Data Records, 30th April 2024. https://eocis.org/wp-content/uploads/2024/06/UK_EOCIS_Quick-Start-Guide-V1.3-Global-LST.pdf*	*No*
ATBD	*Ghent et al., 2025, Land Surface Temperature Climate Change Initiative Algorithm Theoretical Basis Document, Ref: LST-CCI-D2.2-ATBD.* https://admin.climate.esa.int/media/documents/LST-CCI-D2.2-ATBD_-_i4r2_-_Algorithm_Theoretical_Basis_Document.pdf	No

4 Metrology

Sensor Calibration & Characterisation	
Grade: Ideal	
Justification	*Publications describing the instrument calibration, uncertainty budget, and uncertainty validation from mission tandem phase.*
References	• *Smith, D. L., et al. (2014). "Calibration approach and plan for the sea and land surface temperature radiometer." Journal of Applied Remote Sensing, 8(1), 084980.DOI: 10.1117/1.JRS.8.084980* • *Smith D, Hunt SE, Etxaluze M, Peters D, Nightingale T, Mittaz J, Woolliams ER, Polehampton E. "Traceability of the Sentinel-3 SLSTR Level-1 Infrared Radiometric Processing". Remote Sensing. 2021; 13(3):374. https://doi.org/10.3390/rs13030374* • *Hunt, S.E.; Mittaz, J.P.D.; Smith, D.; Polehampton, E.; Yemelyanova, R.; Woolliams, E.R.; Donlon, C. Comparison of the Sentinel-3A and B SLSTR Tandem Phase Data Using Metrological Principles. Remote Sens. 2020, 12, 2893. https://doi.org/10.3390/rs12182893*

Geometric Calibration & Characterisation	
Grade: Ideal	
Justification	*Geometric calibration approach published*
References	• *Smith, D. L., et al. (2014). "Calibration approach and plan for the sea and land surface temperature radiometer." Journal of Applied Remote Sensing, 8(1), 084980.DOI: 10.1117/1.JRS.8.084980*

Metrological Traceability Documentation	
Grade: Good	
Justification	*Level-1 only traceability*
References	• *Smith, D.; Hunt, S.E.; Etxaluze, M.; Peters, D.; Nightingale, T.; Mittaz, J.; Woolliams, E.R.; Polehampton, E. Traceability of the Sentinel-3 SLSTR Level-1 Infrared Radiometric Processing.*

	Remote Sens. 2021, 13, 374. https://doi.org/10.3390/rs13030374

Uncertainty Characterisation	
Grade: Excellent	
Justification	*Publication of approach for all uncertainty components*
References	• *Ghent, D., Veal, K., Trent, T., Dodd, E., Sembhi, H., and Remedios, J. (2019). A New Approach to Defining Uncertainties for MODIS Land Surface Temperature. Remote Sensing, 11, 1021. doi: 10.3390/rs11091021*

5 Product Generation

Calibration Algorithm	
Grade: Excellent	
Justification	*As for the pre-flight calibration section – publications available describing the instrument calibration, uncertainty budget, and uncertainty validation from mission tandem phase.*
References	• *Smith, D. L., et al. (2014). "Calibration approach and plan for the sea and land surface temperature radiometer." Journal of Applied Remote Sensing, 8(1), 084980.DOI: 10.1117/1.JRS.8.084980* • *Smith D, Hunt SE, Etxaluze M, Peters D, Nightingale T, Mittaz J, Woolliams ER, Polehampton E. "Traceability of the Sentinel-3 SLSTR Level-1 Infrared Radiometric Processing". Remote Sensing. 2021; 13(3):374. https://doi.org/10.3390/rs13030374* • *Hunt, S.E.; Mittaz, J.P.D.; Smith, D.; Polehampton, E.; Yemelyanova, R.; Woolliams, E.R.; Donlon, C. Comparison of the Sentinel-3A and B SLSTR Tandem Phase Data Using Metrological Principles. Remote Sens. 2020, 12, 2893.* https://doi.org/10.3390/rs12182893

Geometric Processing	
Grade: Excellent	
Justification	*Geometric calibration approach published*
References	• *Smith, D. L., et al. (2014). "Calibration approach and plan for the sea and land surface temperature radiometer." Journal of Applied Remote Sensing, 8(1), 084980.DOI: 10.1117/1.JRS.8.084980*

Retrieval Algorithm	
Grade: Excellent	
Justification	*Strong heritage of the algorithm, comprehensively validated*
References	• *Ghent, D. J., Anand, J. S., Veal, K., Remedios, J. J. (2024). The operational and climate land surface temperature products from the Sea and Land Surface Temperature Radiometers on Sentinel-*

	3A and 3B. Remote Sensing, 16, 3403. https://doi.org/10.3390/rs16183403

Mission-Specific Processing	
Grade: Excellent	
Cloud masking	
Justification	*Long heritage of published cloud masking algorithm*
Reference	• *Bulgin, C. E., Maidment, R. I., Ghent, D., Merchant, C. J. (2024). Stability of cloud detection methods for Land Surface Temperature (LST) Climate Data Records (CDRs). Remote Sensing of Environment, 315, 114440*